

Skills, Tasks and Technologies: Implications for Employment and Earnings

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1. Introduction

Introduction

- This chapter presents an overview of labor market trends.
- It summarizes the canonical model, tractable and conceptually attractive, that has been successful in explaining most of these trends.
- However, some recent trends necessitate a richer framework.
- To this end, the chapter develops a Ricardian model of Roy-type self-selection of workers into tasks.

2. An Overview of Labor Market Trends

2. An overview of labor market trends

2.1 A brief overview of data sources

2.2 The college/high school wage premium

2.3 Real wage levels by skill group

2.4 Overall wage inequality

2.5 Job polarization

A brief overview of data sources

1. March Current Population Survey (March CPS):
Average weekly earnings and employment in past year for FTFY workers, 1963-2008.
2. May/ORC Current Population Survey (MORG CPS):
Hourly wages and employment in survey week for all wage and salary workers, 1973-2009.
3. Census of Populations (Census):
Wages and employment as March CPS for 1% sample of US population in 1960 and 1970 and 5% sample in 1980, 1990, 2000, larger samples suited for detailed occupational analyses.
4. American Community Survey (ACS):
As Census for 2008.

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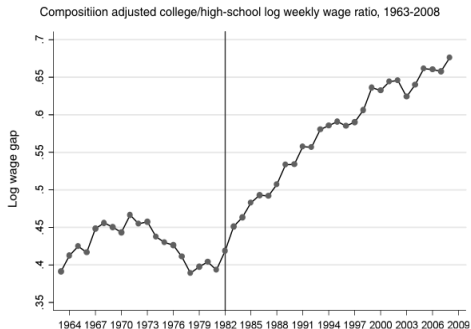


Figure 1 *Source: March CPS data for earnings years 1963-2008. Log weekly wages for full-time, full-year workers are regressed separately by sex in each year on four education dummies (high school dropout, some college, college graduate, greater than college), a quartic in experience, interactions of the education dummies and experience quartic, two race categories (black, non-white other), and a full set of interactions between education, experience, and sex. The composition-adjusted mean log wage is the predicted log wage evaluated for whites at the relevant experience level (5, 15, 25, 35, 45 years) and relevant education level (high school dropout, high school graduate, some college, college graduate, greater than college). The mean log wage for college and high school is the weighted average of the relevant composition adjusted cells using a fixed set of weights equal to the average employment share of each sex by potential experience group. The ratio of mean log wages for college and high school graduates for each year is plotted. See the Data Appendix for more details on the treatment of March CPS data.*

The college/high school wage premium (Figure 1)

1. In 2008, the college premium stood at a maximum 68 log points (or $\exp(0.68) - 1 \approx 0.97$ or 97%).
2. The college premium declined between 1971 and 1978 and increased afterwards.
3. This is critical to our understanding of supply and demand in the determination of between-group wage inequality.

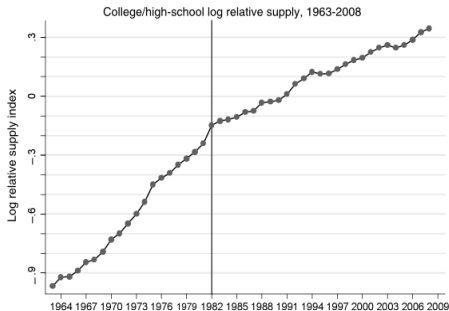


Figure 2 Source: March CPS data for earnings years 1963-2008. Labor supply is calculated using all persons aged 16-64 who reported having worked at least one week in the earnings years, excluding those in the military. The data are sorted into sex-education-experience groups of two sexes (male/female), five education groups (high school dropout, high school graduate, some college, college graduate, and greater than college) and 49 experience groups (0-48 years of potential experience). The number of years of potential experience is calculated by subtracting the number six (the age at which one begins school) and the number of years of schooling from the age of the individual. This number is further adjusted using the assumption that an individual cannot begin work before age 16 and that experience is always non-negative. The labor supply for college/high school groups by experience level is calculated using efficiency units, equal to mean labor supply for broad college (including college graduates and greater than college) and high school (including high school dropouts and high school graduate) categories, weighted by fixed relative average wage weights for each cell. The labor supply of the "some college" category is allocated equally between the broad college and high school categories. The fixed set of wage weights for 1963-2008 are constructed using the average wage in each of the 490 cells (2 sexes, 5 education groups, 49 experience groups) over this time period.

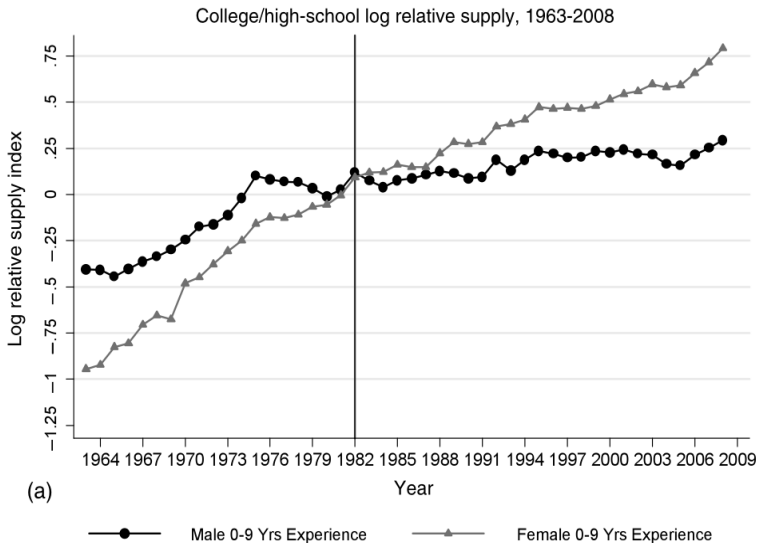
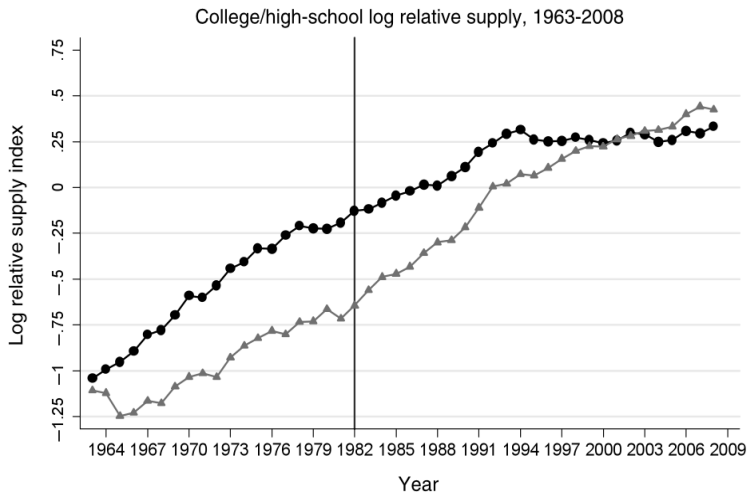


Figure 3 Source: March CPS data for earnings years 1963-2008. See note to [Fig. 2](#). Log relative supply for 0-9 and 20-29 years of potential experience is plotted for males and females.



(b) —●— Male 20-29 Yrs Experience —▲— Female 20-29 Yrs Experience

Figure 3 Source: March CPS data for earnings years 1963-2008. See note to Fig. 2. Log relative supply for 0-9 and 20-29 years of potential experience is plotted for males and females.

Slowing growth in college attainment (Figures 2 & 3)

1. The Vietnam War boosted college attainment in the late 1960s and early 1970s to defer military service.
2. Large baby boom cohorts that entered the labor market in the 1960s and 1970s were both more educated and more numerous than existing cohorts.
3. The decline in the college premium in the 1970s (temporarily) discouraged college enrollment.
4. The male college completion rate never returned to its pre-1975 trajectory.

2. An overview of labor market trends

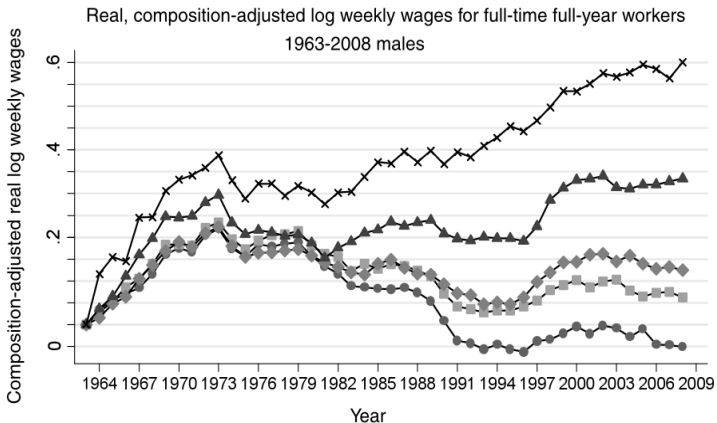
2.1 A brief overview of data sources

2.2 The college/high school wage premium

2.3 Real wage levels by skill group

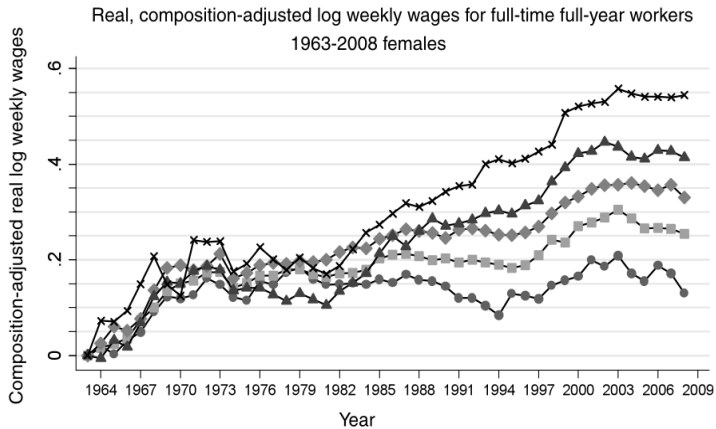
2.4 Overall wage inequality

2.5 Job polarization



(a)

Figure 4 Source: March CPS data for earnings years 1963-2008. See note to Fig. 1. The real log weekly wage for each education group is the weighted average of the relevant composition adjusted cells using a fixed set of weights equal to the average employment share of each group. Nominal wage values are deflated using the Personal Consumption Expenditure (PCE) deflator.



(b)

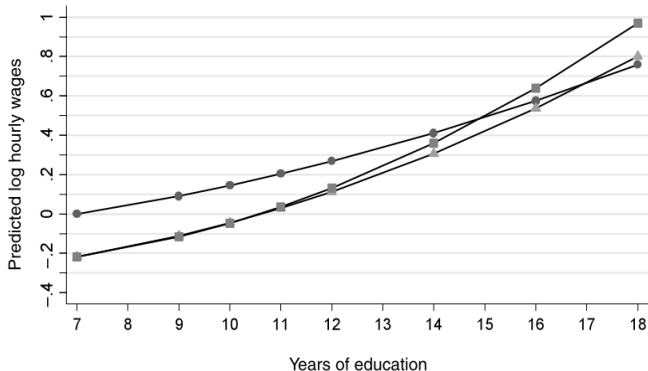
Figure 4 Source: March CPS data for earnings years 1963-2008. See note to Fig. 1. The real log weekly wage for each education group is the weighted average of the relevant composition adjusted cells using a fixed set of weights equal to the average employment share of each group. Nominal wage values are deflated using the Personal Consumption Expenditure (PCE) deflator.

Real wage levels by skill group (Figure 4)

- 1963-1973: Real wages rose steeply and uniformly for both genders and all education groups.
- 1973-1980: Wage levels stagnated for both genders and all education groups.
- After 1980: Real wage levels of less-educated (less-experienced men) fell, whereas real wages for post-college educated rose.
- Earnings inequality increased rapidly in the 1980s and 1990s (especially for men) and modestly in the 2000s.

Predicted log hourly wages by years of education, education quadratic:

Males



(a)

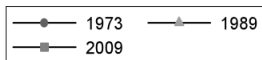
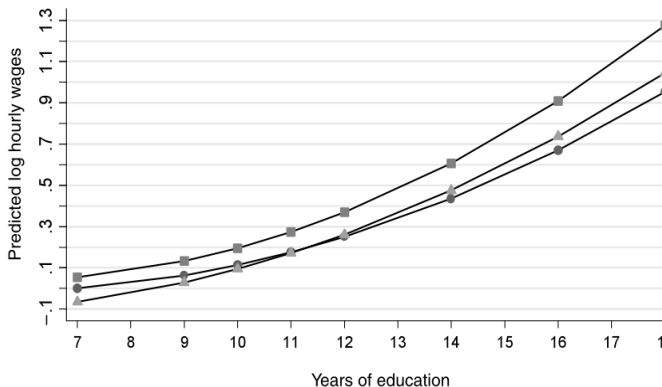


Figure 5 Source: May/ORG CPS data for earnings years 1973-2009. For each year, log hourly wages for all workers, excluding the self-employed and those employed by the military, are regressed on a quadratic in education (eight categories), a quartic in experience, a female dummy, and interactions of the female dummy and the quartic in experience. Predicted real log hourly wages are computed in 1973, 1989 and 2009 for each of the years of schooling presented in the figure. See the Data Appendix for more details on the treatment of May/ORG CPS data.

Predicted log hourly wages by years of education, education quadratic:

Females



(b)

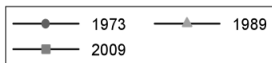
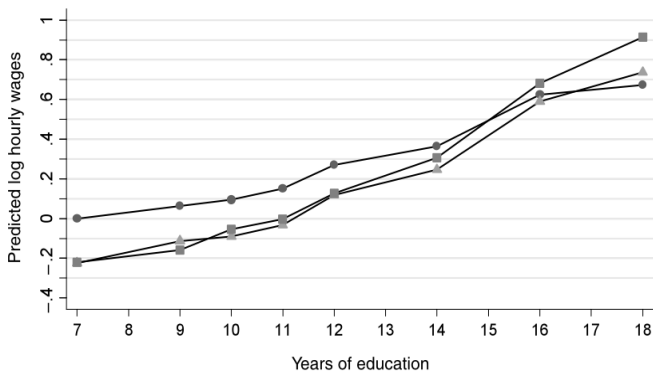


Figure 5 Source: May/ORG CPS data for earnings years 1973-2009. For each year, log hourly wages for all workers, excluding the self-employed and those employed by the military, are regressed on a quadratic in education (eight categories), a quartic in experience, a female dummy, and interactions of the female dummy and the quartic in experience. Predicted real log hourly wages are computed in 1973, 1989 and 2009 for each of the years of schooling presented in the figure. See the Data Appendix for more details on the treatment of May/ORG CPS data.

Predicted log hourly wages by years of education, education dummies:

Males



(a)

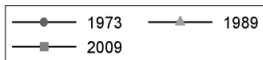
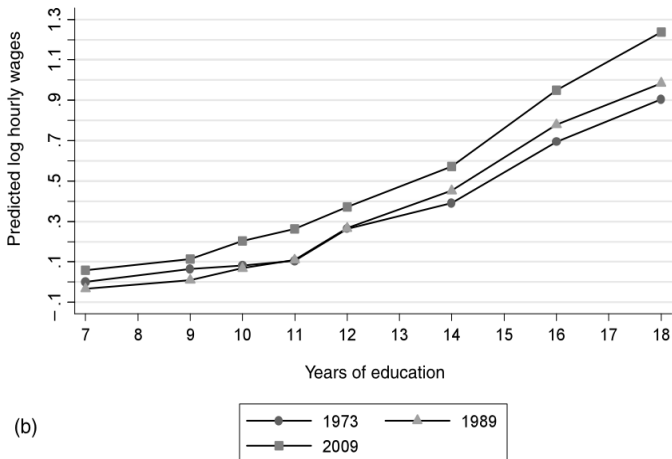


Figure 6 Source: May/ORG CPS data for earnings years 1973-2009. For each year, log hourly wages for all workers, excluding the self-employed and those employed by the military, are regressed on eight education dummies, a quartic in experience, a female dummy, and interactions of the female dummy and the quartic in experience. Predicted real log hourly wages are computed in 1973, 1989 and 2009 for each of the years of schooling presented. See the Data Appendix for more details on the treatment of May/ORG CPS data.

Predicted log hourly wages by years of education, education dummies:

Females



(b)

Figure 6 Source: May/ORG CPS data for earnings years 1973-2009. For each year, log hourly wages for all workers, excluding the self-employed and those employed by the military, are regressed on eight education dummies, a quartic in experience, a female dummy, and interactions of the female dummy and the quartic in experience. Predicted real log hourly wages are computed in 1973, 1989 and 2009 for each of the years of schooling presented. See the Data Appendix for more details on the treatment of May/ORG CPS data.

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Overall wage inequality

- Evolution of real and relative wages by education, gender and experience groups does not capture all changes in the wage distribution.
- Summarize the entire earnings distribution by showing trends in real wages by percentile.
- Focus on the 10th to 90th percentiles because CPS samples are unlikely to provide accurate measures of earnings at the highest and lowest percentiles.

Cumulative log change in real weekly earnings at the 90th, 50th and 10th
wage percentiles

1963-2008: full-time full-year males and females

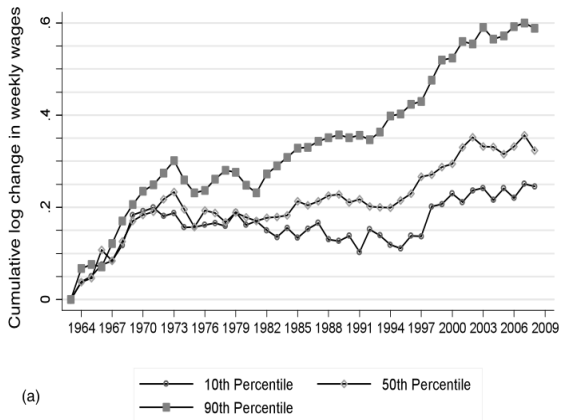


Figure 7 Source: March CPS data for earnings years 1963-2008. For each year, the 10th, median and 90th percentiles of log weekly wages are calculated for full-time, full-year workers.

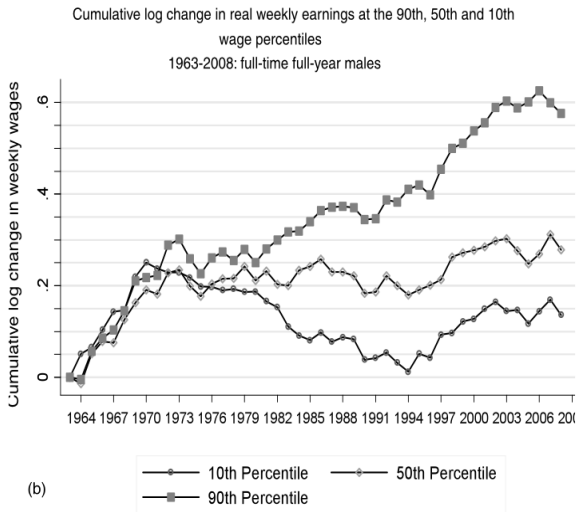


Figure 7 Source: March CPS data for earnings years 1963-2008. For each year, the 10th, median and 90th percentiles of log weekly wages are calculated for full-time, full-year workers.

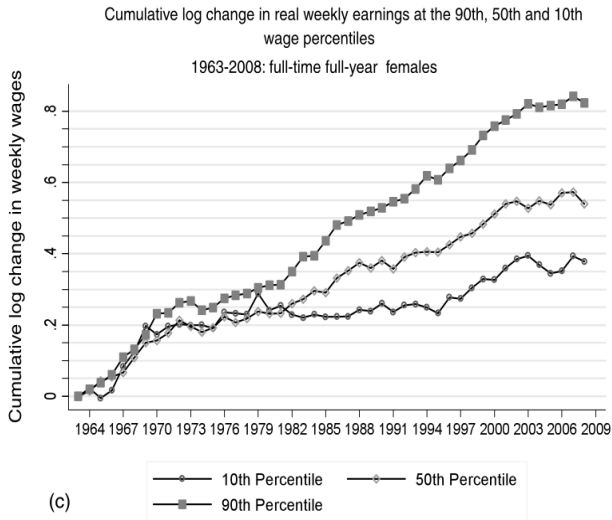


Figure 7 Source: March CPS data for earnings years 1963-2008. For each year, the 10th, median and 90th percentiles of log weekly wages are calculated for full-time, full-year workers.

Overall wage inequality (Figure 7)

- 1963-1973: All percentiles rise relatively evenly.
- 1973-1980: Median and 10th percentile stagnate, and the 90th percentile pulls away modestly.
- 1980-2000: Median is flat, the 10th percentile falls and the 90th percentile grows, increasing overall inequality.
- After 2000: Median and 10th percentile grow and the 90th percentile grows faster, modestly increasing overall inequality.

Cumulative log change in real hourly earnings at the 90th, 50th and 10th wage percentiles
1974-2008: males and females

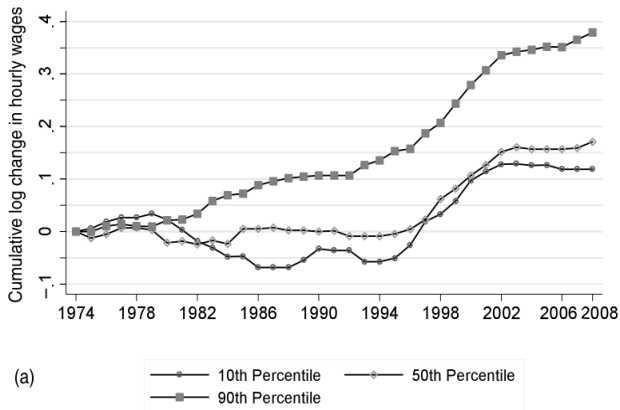


Figure 8 Source: May/ORG CPS data for earnings years 1973-2009. The data are pooled using three-year moving averages (i.e. the year 1974 includes data from years 1973, 1974 and 1975). For each year, the 10th, median and 90th percentiles of log weekly wages are calculated for all workers, excluding the self-employed and those employed in military occupations.

Cumulative log change in real hourly earnings at the 90th, 50th and 10th wage percentiles
1974-2008: males

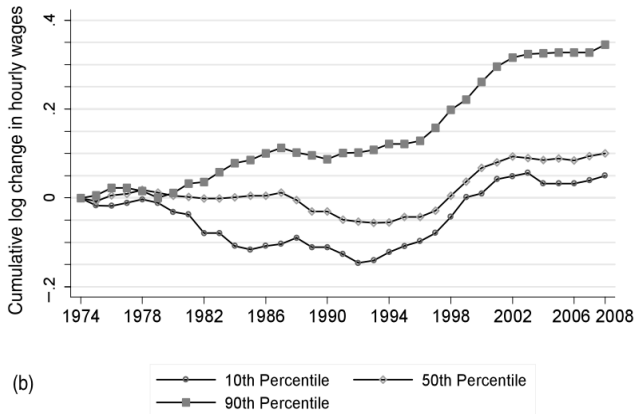
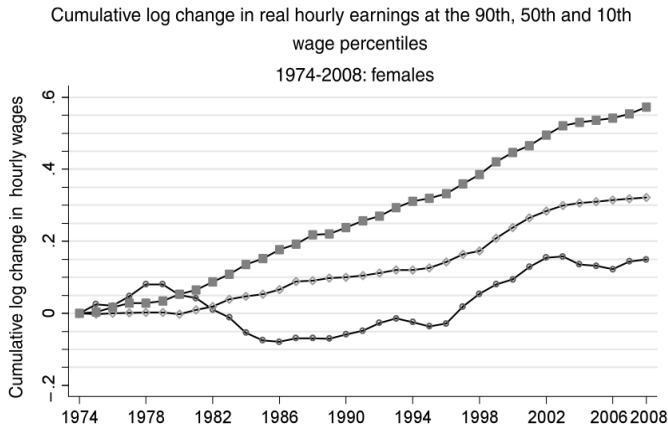


Figure 8 Source: May/ORG CPS data for earnings years 1973-2009. The data are pooled using three-year moving averages (i.e. the year 1974 includes data from years 1973, 1974 and 1975). For each year, the 10th, median and 90th percentiles of log weekly wages are calculated for all workers, excluding the self-employed and those employed in military occupations.



(c)

—○— 10th Percentile —◇— 50th Percentile
 —■— 90th Percentile

Figure 8 Source: May/ORG CPS data for earnings years 1973-2009. The data are pooled using three-year moving averages (i.e. the year 1974 includes data from years 1973, 1974 and 1975). For each year, the 10th, median and 90th percentiles of log weekly wages are calculated for all workers, excluding the self-employed and those employed in military occupations.

Overall wage inequality (Figure 8)

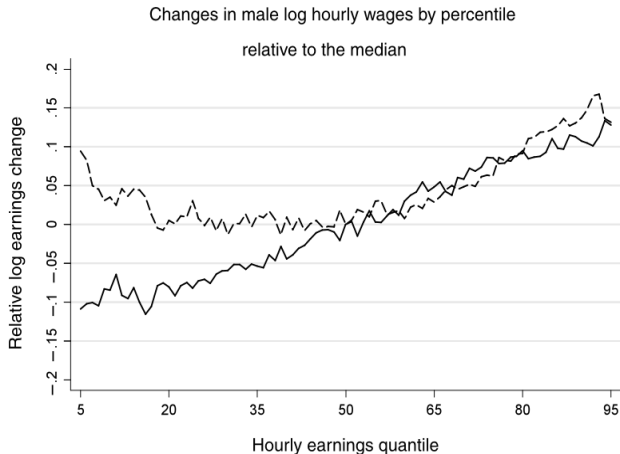
- Figure 7 depicted trends in average weekly wages for for FTFY workers using the March CPS.
- Figure 8 shows trends in average hourly wages for all hourly and salary workers using the MORG CPS.
- The evolution of the 10th percentile is much more volatile: It decreases between 1979-1988, stagnates between 1988-1995, grows rapidly between 1995-2003, and stagnates afterwards.
- There has been “wage polarization” after 1988.



(a)

— 1974-1988 - - - - 1988-2008

Figure 9 Source: May/ORG CPS data for earnings years 1973-2009. The data are pooled using three-year moving averages (i.e. the year 1974 includes data from years 1973, 1974 and 1975). For each year, the 5th through 95th percentiles of log hourly wages are calculated for all workers, excluding the self-employed and those employed in military occupations. The log wage change at the median is normalized to zero in each time interval.



(b)

— 1974-1988 - - - - 1988-2008

Figure 9 Source: May/ORG CPS data for earnings years 1973-2009. The data are pooled using three-year moving averages (i.e. the year 1974 includes data from years 1973, 1974 and 1975). For each year, the 5th through 95th percentiles of log hourly wages are calculated for all workers, excluding the self-employed and those employed in military occupations. The log wage change at the median is normalized to zero in each time interval.

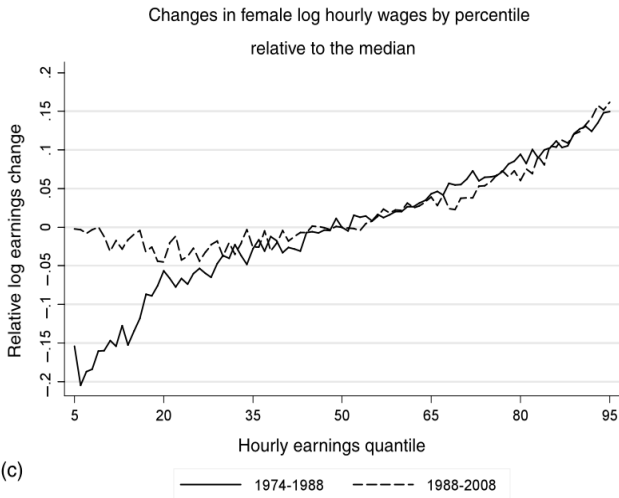


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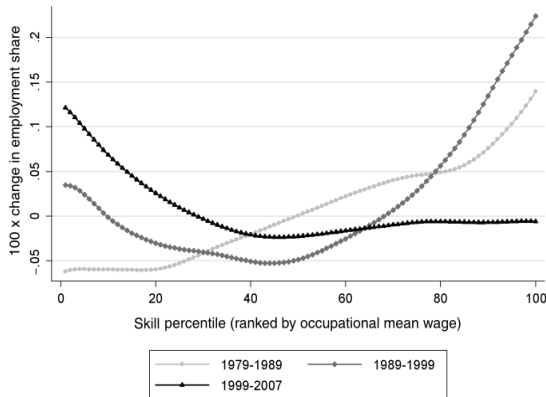


Figure 10 Source: Census IPUMS 5 percent samples for years 1980, 1990, and 2000, and Census American Community Survey for 2008. All occupation and earnings measures in these samples refer to prior year's employment. The figure plots log changes in employment shares by 1980 occupational skill percentile rank using a locally weighted smoothing regression (bandwidth 0.8 with 100 observations), where skill percentiles are measured as the employment-weighted percentile rank of an occupation's mean log wage in the Census IPUMS 1980 5 percent extract. The mean log wage in each occupation is calculated using workers' hours of annual labor supply times the Census sampling weights. Consistent occupation codes for Census years 1980, 1990, and 2000, and 2008 are from [Autor and Dorn \(2009\)](#).

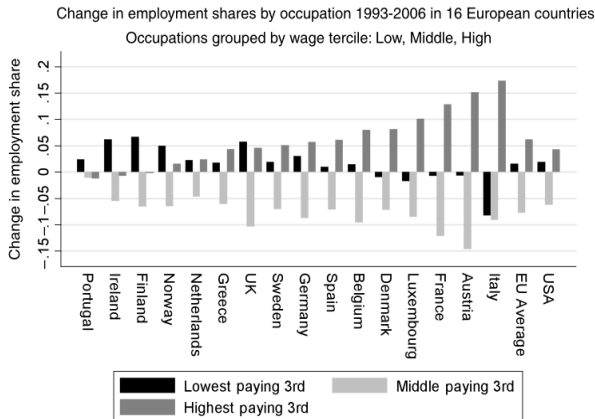


Figure 11 Source: Data on EU employment are from [Goos et al. \(2009\)](#). US data are from the May/ORG CPS files for years 1993-2006. The data include all persons aged 16-64 who reported employment in the sample reference week, excluding those employed by the military and in agricultural occupations. Occupations are first assigned to 326 occupation groups that are consistent over the given time period. These occupations are then grouped into three broad categories by wage level.

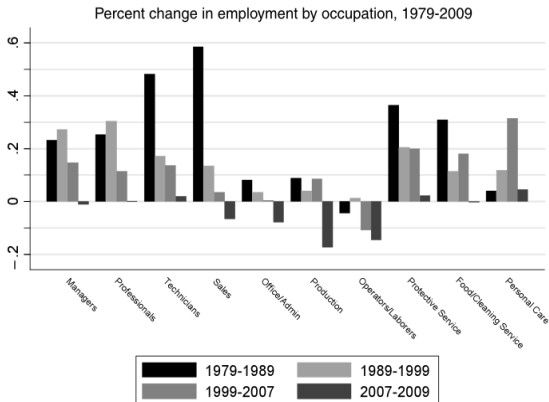


Figure 12 *Source: May/ORG CPS files for earnings years 1979-2009. The data include all persons aged 16-64 who reported employment in the sample reference week, excluding those employed by the military and in agricultural occupations. Occupations are assigned to 326 occupation groups that are consistent over the given time period. All non-military, non-agricultural occupations are assigned to one of ten broad occupations presented in the figure.*

Employment shares by major occupation groups, 1959-2007:

Males and females

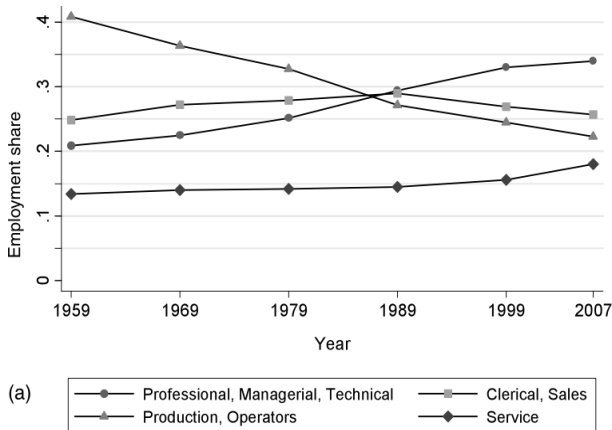
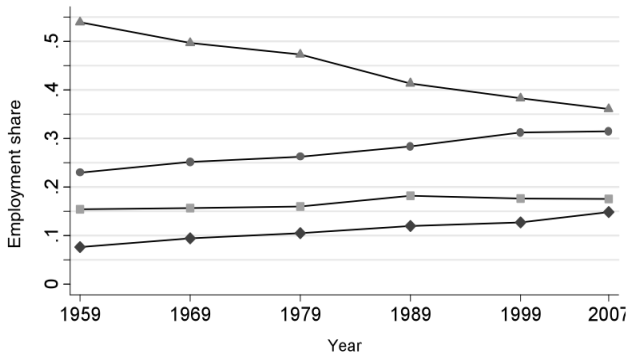


Figure 13 Source: Census IPUMS 5 percent samples for years 1960, 1970, 1980, 1990, and 2000, and Census American Community Survey for 2008. The data include all persons aged 16-64 who reported having worked last year, excluding those employed by the military and in agricultural occupations. Occupations are first assigned to 326 occupation groups that are consistent over the given time period. All non-military, non-agricultural occupations are assigned to one of four broad occupations.

Employment shares by major occupation groups, 1959-2007:

Males



(b)

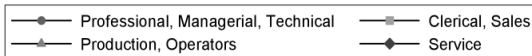


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Employment shares by major occupation groups, 1959-2007:

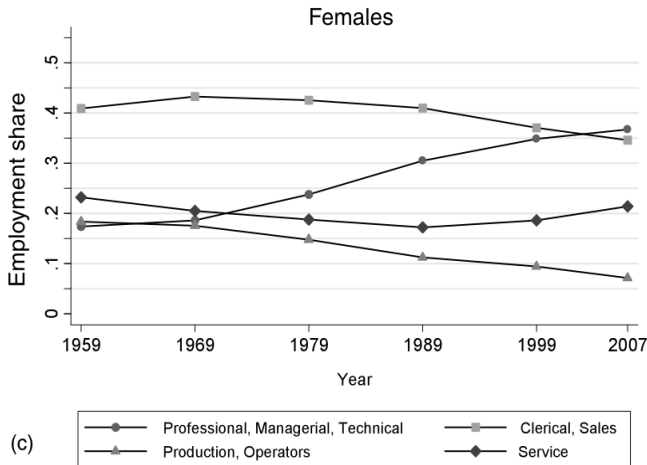


Figure 13 Source: Census IPUMS 5 percent samples for years 1960, 1970, 1980, 1990, and 2000, and Census American Community Survey for 2008. The data include all persons aged 16-64 who reported having worked last year, excluding those employed by the military and in agricultural occupations. Occupations are first assigned to 326 occupation groups that are consistent over the given time period. All non-military, non-agricultural occupations are assigned to one of four broad occupations.

Is job polarization explained by industrial composition?

- The change in employment in occupation j can be decomposed as:

$$\Delta E_{jt} = \sum_k \Delta E_{kt} \lambda_{jk} + \sum_k \Delta \lambda_{jkt} E_k \quad (1)$$

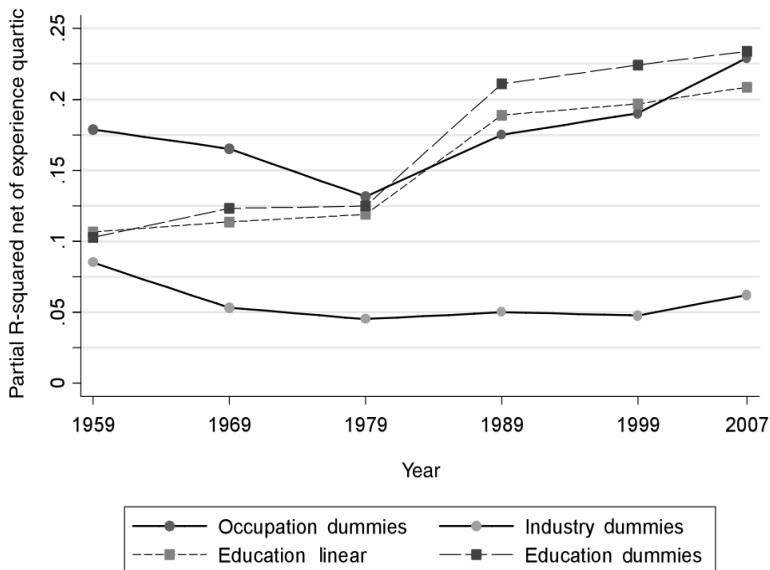
with ΔE_{kt} the change over time in employment in industry k ; λ_{jk} the share of occupation j in industry k averaged across years; $\Delta \lambda_{jkt}$ the change over time in the share of occupation j in industry k ; and E_k employment in k averaged over time.

- Within-industry shifts against middle-skilled and favoring high- and low-skilled occupations are the primary driver for job polarization.

The growing importance of occupations in wages

- Employment polarization combined with wage polarization suggests that workers' occupational affiliations may have become a more important determinant of wages.
- Use Census and ACS data for 1959-2007 and regress FTFY weekly wages onto education, experience, occupation dummies, industry dummies, and task measures.
- Calculate the partial R^2 (net of experience) for each set of regressors in each year for men and women separately.
- Partial R^2 for occupation dummies and task measures have increased since 1979.

Males: Partial R-squared net of experience quartic, 1959-2007



Females: Partial R-squared net of experience quartic, 1959-2007

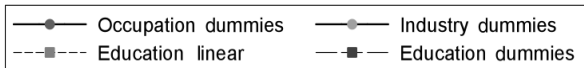
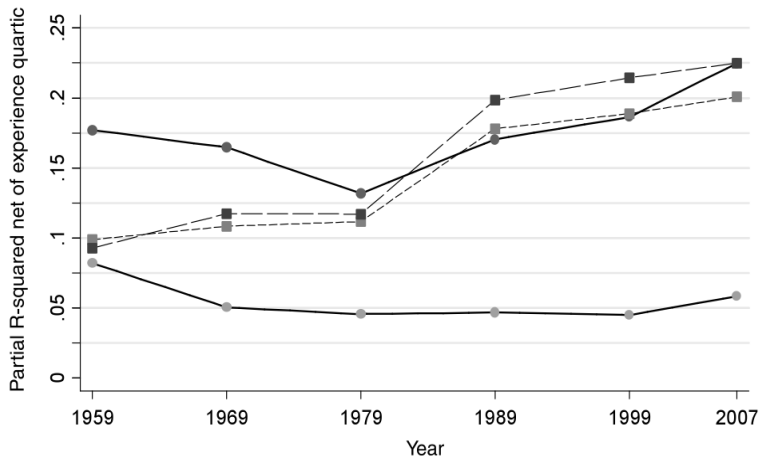
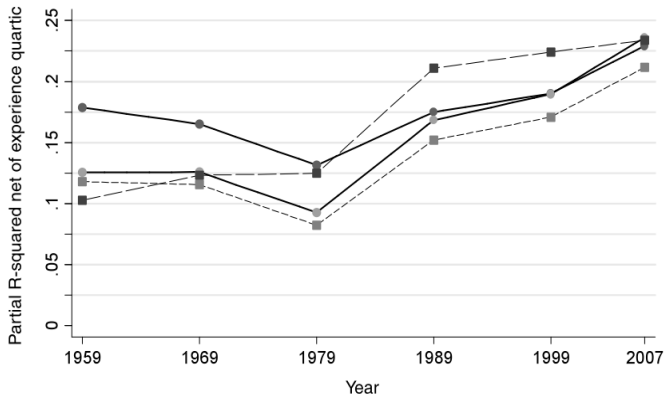


Figure 17 Source: Census IPUMS 5 percent samples for years 1960, 1970, 1980, 1990, and 2000, and Census American Community Survey for 2008. The data include all full-time, full-year workers aged 16-64, excluding those employed by the military and in agricultural occupations. Linear education measure is equal to years of educational attainment. For those who have not completed second grade, their years of education are imputed based on gender and ethnicity. Education dummies consist of five broad categories: high school dropouts, high school graduates, some college education, college graduates, and post-college degree. Occupations are assigned to 326 occupation groups that are consistent over the given time period. From these groups, occupations are then consolidated into ten broad categories: Managers; Professionals; Technicians; Sales; Office and administrative; Production, craft and repair; Operators, fabricators and laborers; Protective service; Food prep, buildings and grounds, cleaning; and Personal care and personal services. Industries are similarly converted from their respective scheme to a consistent set of 149 industries, as used in [Autor et al. \(1998\)](#). From these 149 industries, ten broad industry categories are constructed and include: Construction; Manufacturing; Transport and utilities; Wholesale trade; Retail trade; Finance, Insurance and Real Estate; Business services; Personal services and entertainment; Professional services; and Public administration. The partial R-squared values presented above are calculated as follows: Log weekly wages and each variable group above are orthogonalized using a quartic in experience and two ethnicity dummies. Using the residuals from each of these regressions, residual log weekly wages are regressed separately on the residuals from the variable groups of interest, and the R-squared value from this regression is plotted above for each year. All regressions are weighted by Census person weights.

Males: Partial R-squared net of experience quartic, 1959-2007



(a)

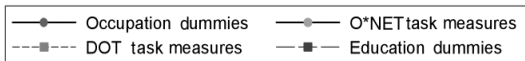
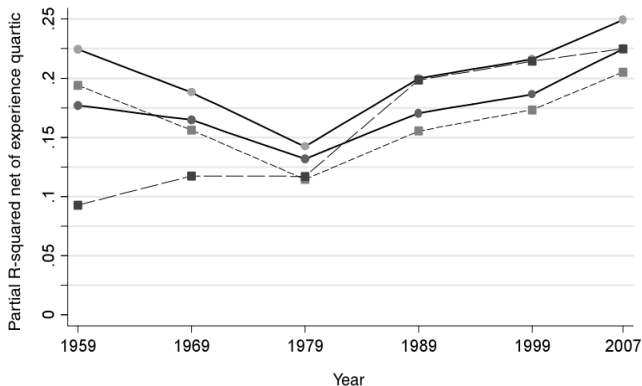


Figure 18 Source: Census IPUMS 5 percent samples for years 1960, 1970, 1980, 1990, and 2000, and Census American Community Survey for 2008. See note to Fig. 17 for the partial R-squared calculation procedure. Five O*NET constructed task measures, constructed from a combination of O*NET activities and context scores, are utilized: routine cognitive, routine manual, non-routine cognitive analytic, non-routine manual, and non-routine interpersonal. Three DOT task measures are utilized, as in Autor et al. (2003): abstract, routine, and manual. See the Data Appendix for more information on the construction of the O*NET task measures.

Females: Partial R-squared net of experience quartic, 1959-2007



(b)

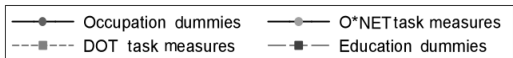


Figure 18 Source: Census IPUMS 5 percent samples for years 1960, 1970, 1980, 1990, and 2000, and Census American Community Survey for 2008. See note to Fig. 17 for the partial R-squared calculation procedure. Five O*NET constructed task measures, constructed from a combination of O*NET activities and context scores, are utilized: routine cognitive, routine manual, non-routine cognitive analytic, non-routine manual, and non-routine interpersonal. Three DOT task measures are utilized, as in Autor et al. (2003): abstract, routine, and manual. See the Data Appendix for more information on the construction of the O*NET task measures.

3. The Canonical Model

3. The canonical model

3.1 The simple theory of the canonical model

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Tinbergen's race between education and technology

Jan Tinbergen, 1975

"The two preponderant forces at work are technological development, which made for a relative increase in demand and hence in the income ratio... and increased access to schooling, which made for a relative decrease."

Translation into the "canonical model":

- Long term trend increases towards greater relative demand and greater supply of skilled workers.
- Bursts of supply and/or technologically-induced demand accelerations/decelerations that cause demand to temporarily move out more rapidly than supply or vice versa in some eras.

CES model

- Consider an aggregate CES function:

$$Y = \left[[A_L L]^{\frac{\sigma-1}{\sigma}} + [A_H H]^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} \quad (2)$$

with $L = \int_{i \in L} l_i di$ and $H = \int_{i \in H} h_i di$ where l_i (h_i) denotes the supply in efficiency units of low (high) educated labor that worker i possesses and with $\sigma \in [0, \infty)$ the elasticity of substitution between L and H .

- Changes in H/L will capture changes in the relative supply of educated labor, whereas all other changes will capture changes in the relative demand for educated labor.

Interpretations of the CES model

1. Only one good, skilled and unskilled workers are imperfect substitutes in its production.
2. Two-good economy:
 - Consumers have CES utility with elasticity of substitution in consumption of σ
 - Good Y_L is produced with $Y_L = A_L L$
 - Good Y_H is produced with $Y_H = A_H H$
3. A mixture of these two where different sectors produce goods that are imperfect substitutes, and low and high skilled workers are employed in all sectors.

Interpretations of σ

- The CES model is an abstraction.
- σ combines substitution in production and consumption across firms, industries, consumers,...
- Surprising degree of consensus that $\sigma \in [1, 2]$
- When examining the impact of technological progress, we assume that σ captures the elasticity of substitution between input factors in an aggregate production function.

Technological change in the CES model

- Consider a more general aggregate CES production function:

$$Y = \Pi \left[\Gamma [A_L L + B_L]^{\frac{\sigma-1}{\sigma}} + [1 - \Gamma] [A_H H + B_H]^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$$

- Sources of technological change are:
 1. A_L and A_H : Augmenting the productivity of factors (“intensive” margin technological change).
 2. Γ : The distribution parameter reallocates production across factors (“extensive” margin technological change).
 3. Π : Hicks-neutral TFP.
 4. B_L and B_H : Directly labor-replacing technologies.

Wage setting

- Assume that technological change is only factor-augmenting and skill-biased, $A_H/A_L \uparrow$ or SBTC.
- Given the aggregate CES production function:

$$Y = \left[[A_L L]^{\frac{\sigma-1}{\sigma}} + [A_H H]^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} \quad (2)$$

- In equilibrium, productive efficiency implies:

$$w_L = \frac{\partial Y}{\partial L} = A_L^{\frac{\sigma-1}{\sigma}} \left[A_L^{\frac{\sigma-1}{\sigma}} + A_H^{\frac{\sigma-1}{\sigma}} [H/L]^{\frac{\sigma-1}{\sigma}} \right]^{\frac{1}{\sigma-1}} \quad (3)$$

$$w_H = \frac{\partial Y}{\partial H} = A_H^{\frac{\sigma-1}{\sigma}} \left[A_L^{\frac{\sigma-1}{\sigma}} [H/L]^{-\frac{\sigma-1}{\sigma}} + A_H^{\frac{\sigma-1}{\sigma}} \right]^{\frac{1}{\sigma-1}} \quad (4)$$

Comparative statics for low-skilled wages

- We had that:

$$w_L = \frac{\partial Y}{\partial L} = A_L^{\frac{\sigma-1}{\sigma}} \left[A_L^{\frac{\sigma-1}{\sigma}} + A_H^{\frac{\sigma-1}{\sigma}} [H/L]^{\frac{\sigma-1}{\sigma}} \right]^{\frac{1}{\sigma-1}} \quad (3)$$

- This implies the following comparative statics:

1. $\frac{\partial w_L}{\partial L} < 0$ because of decreasing marginal productivity in L
2. $\frac{\partial w_L}{\partial H} > 0$ because of q-complementarity
3. $\frac{\partial w_L}{\partial A_L} > 0 \Leftrightarrow \sigma \geq 1$ ($\Leftrightarrow \sigma > 1 - s_L$ with s_L wagebillshare of L)
4. $\frac{\partial w_L}{\partial A_H} > 0$ because of q-complementarity

Comparative statics for high-skilled wages

- We had that:

$$w_H = \frac{\partial Y}{\partial H} = A_H^{\frac{\sigma-1}{\sigma}} \left[A_L^{\frac{\sigma-1}{\sigma}} [H/L]^{-\frac{\sigma-1}{\sigma}} + A_H^{\frac{\sigma-1}{\sigma}} \right]^{\frac{1}{\sigma-1}} \quad (4)$$

- This implies the following comparative statics:

1. $\frac{\partial w_H}{\partial H} < 0$ because of decreasing marginal productivity in H
2. $\frac{\partial w_H}{\partial L} > 0$ because of q-complementarity
3. $\frac{\partial w_H}{\partial A_H} > 0 \Leftrightarrow \sigma \geq 1$ ($\Leftrightarrow \sigma > 1 - s_H = s_L$)
4. $\frac{\partial w_H}{\partial A_L} > 0$ because of q-complementarity

The skill premium

- Combining (3) and (4), the skill premium is given by:

$$\omega \equiv \frac{w_H}{w_L} = \left[\frac{A_H}{A_L} \right]^{\frac{\sigma-1}{\sigma}} \left[\frac{H}{L} \right]^{-\frac{1}{\sigma}} \quad (5)$$

- Rewriting by taking logs gives:

$$\ln(\omega) = \frac{\sigma-1}{\sigma} \ln \left(\frac{A_H}{A_L} \right) - \frac{1}{\sigma} \ln \left(\frac{H}{L} \right) \quad (6)$$

where the first term captures the first force in Tinbergen's race (i.e. technology) and the second term captures the second force (i.e. education).

Comparative statics for the skill premium

- We had that the skill premium is given by:

$$\ln(\omega) = \frac{\sigma - 1}{\sigma} \ln\left(\frac{A_H}{A_L}\right) - \frac{1}{\sigma} \ln\left(\frac{H}{L}\right) \quad (6)$$

- Equation (6) implies the following comparative statics:

$$\frac{\partial \ln(\omega)}{\partial \ln(H/L)} = -\frac{1}{\sigma} < 0 \quad (7)$$

$$\frac{\partial \ln(\omega)}{\partial \ln(A_H/A_L)} = \frac{\sigma - 1}{\sigma} \geq 0 \text{ if } \sigma \geq 1 \quad (8)$$

The skill premium and skill supplies

- We had the comparative static:

$$\frac{\partial \ln(\omega)}{\partial \ln(H/L)} = -\frac{1}{\sigma} < 0 \quad (7)$$

- There are 2 types of substitution depending on how you interpret the CES model:
 1. H and L are used to produce the same good and an increase in H/L reduces the marginal productivity of H relative to L and hence its relative wage.
 2. H and L are producing different goods and an increase in H/L reduces the marginal utility of consumption of H relative to L and hence the its real price and relative wage.

The skill premium and SBTC

- We had the comparative static:

$$\frac{\partial \ln(\omega)}{\partial \ln(A_H/A_L)} = \frac{\sigma - 1}{\sigma} \geq 0 \text{ if } \sigma \geq 1 \quad (8)$$

- What happens to the skill premium depends on $\sigma \geq 1$:
 1. If $\sigma > 1$: Despite an increase in A_H/A_L , *relative* demand for H/L increases because there is enough substitution towards H .
 2. If $\sigma = 1$: The skill premium doesn't change.
 3. If $\sigma < 1$: Because of an increase in A_H/A_L and limited substitution towards H , *relative* demand for H/L decreases.

Summary of key relationships

An increase in H holding L , A_H and A_L constant:

1. Skilled workers: $w_H \downarrow$ (decreasing marginal product).
2. Unskilled workers: $w_L \uparrow$ (q-complementarity)
3. Skill premium: $\omega \equiv w_H / w_L \downarrow$

An increase in A_H holding A_L , H and L constant:

1. Skilled workers: $w_H \uparrow \Leftrightarrow \sigma \geq 1$ ($\Leftrightarrow \sigma > 1 - s_H = s_L$)
2. Unskilled workers: $w_L \uparrow$ (q-complementarity)
3. Skill premium: $\omega \equiv w_H / w_L \geq 0$ if $\sigma \geq 1$

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Bringing Tinbergen's education race to the data

- Assume that SBTC is log linear in demand for skills:

$$\ln\left(\frac{A_{Ht}}{A_{Lt}}\right) = \gamma_0 + \gamma_1 t \text{ with } \gamma_1 > 0 \quad (9)$$

- Substituting into equation (6) gives:

$$\ln(\omega_t) = \frac{\sigma - 1}{\sigma} \gamma_0 + \frac{\sigma - 1}{\sigma} \gamma_1 t - \frac{1}{\sigma} \ln\left(\frac{H_t}{L_t}\right) \quad (10)$$

- Assuming that $\sigma > 1$, the skill premium will increase if SBTC exceeds the growth in education.

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An update of Katz & Murphy [92]

- Katz & Murphy [92] formalized Tinbergen's race between education and technology.
- They introduced the “canonical model” and “SBTC”.
- For the period 1963-1987, Katz & Murphy [92] estimate:

$$\ln\left(\frac{w_{Ht}}{w_{Lt}}\right) = \beta_0 + \beta_1 \ln\left(\frac{H_t}{L_t}\right) + \beta_2 t + \epsilon_t$$

- Consider the period 1963-1987 as in Katz & Murphy [92].
- Make out of sample predictions for the period 1987-2008.

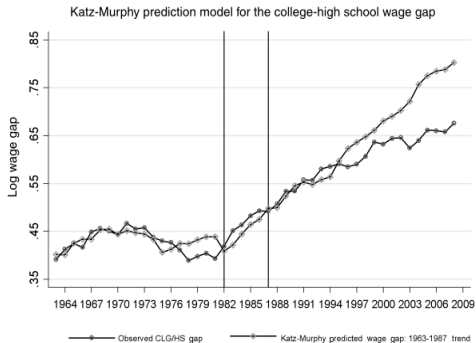


Figure 19 Source: March CPS data for earnings years 1963–2008. Log weekly wages for full-time, full-year workers are regressed separately by sex in each year on four education dummies (high school dropout, some college, college graduate, greater than college), a quartic in experience, interactions of the education dummies and experience quartic, and two race categories (black, non-white other). The composition-adjusted mean log wage is the predicted log wage evaluated for whites at the relevant experience level (5, 15, 25, 35, 45 years) and relevant education level (high school dropout, high school graduate, some college, college graduate, greater than college). The mean log wage for college and high school is the weighted average of the relevant composition adjusted cells using a fixed set of weights equal to the average employment share of each sex by experience group. The ratio of mean log wages for college and high school graduates for each year is plotted. See the Data Appendix for more details on the treatment of March CPS data. The Katz-Murphy predicted wage gap series contains the predicted values from a regression of the college/high school wage gap on time trend term and log labor supply, as measured in efficiency units described in the note to [Fig. 2](#), for years 1963–1987.

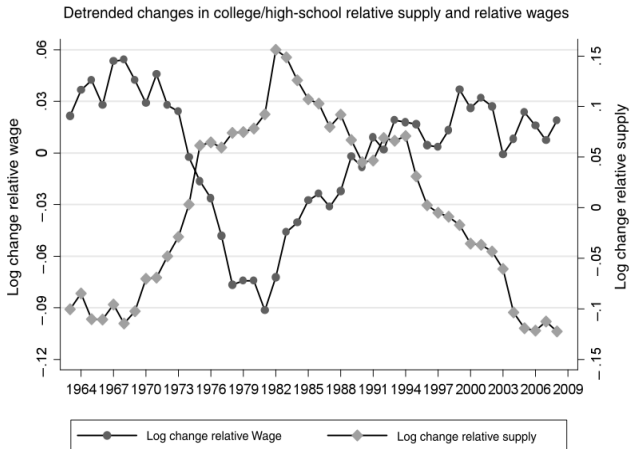


Figure 20 *Source: March CPS data for earnings years 1963–2008. See note to [Fig. 19](#). The detrended supply and wage series are the residuals from separate OLS regressions of the relative supply and relative wage measures on a constant and a linear time trend.*

Table 8 Regression models for the college/high school log wage gap, 1963-2008.

	1963-1987		1963-2008		
	(1)	(2)	(3)	(4)	(5)
CLG/HS relative supply	-0.612 (0.128)	-0.339 (0.043)	-0.644 (0.066)	-0.562 (0.112)	-0.556 (0.094)
Time	0.027 (0.005)	0.016 (0.001)	0.028 (0.002)	0.029 (0.006)	0.020 (0.006)
Time X post-1992			-0.010 (0.002)		
Time ² /100				-0.013 (0.006)	0.036 (0.012)
Time ³ /1000					-0.007 (0.002)
Constant	-0.217 (0.134)	0.059 (0.039)	-0.254 (0.066)	-0.189 (0.122)	-0.145 (0.103)
Observations	25	46	46	46	46
R-squared	0.558	0.935	0.961	0.941	0.960

Source: March CPS data for earnings years 1963-2008. See notes to [Figs 2](#) and [19](#).

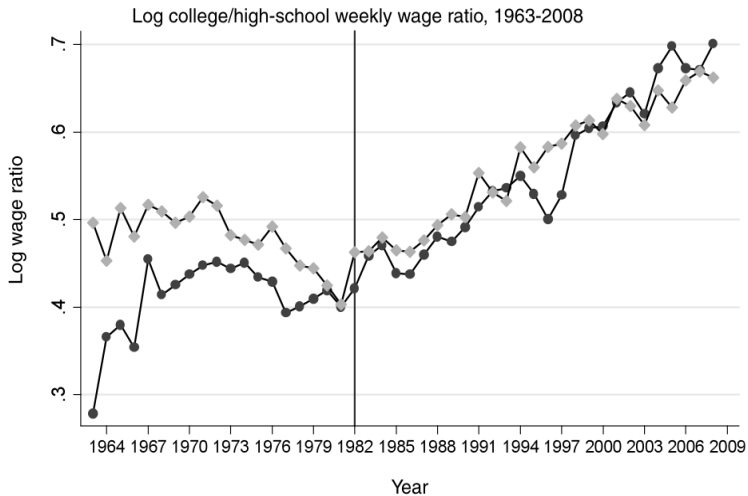
Exploiting a richer set of facts

- Additional identification and explanatory power can come from considering a richer set of facts.
- Changes in the college premium were different by age groups: The college premium increased faster for younger workers in the early 1980s and for older workers in the late 1990s.
- This could be because cohorts entering the labor market in the 1980s were relatively less educated and aged thereafter.

Log college/high-school weekly wage ratio, 1963-2008



Figure 21 Source: March CPS data for earnings years 1963-2008. See note to Fig. 19. Log college/high school weekly wage ratio for 0-9 and 20-29 years of potential experience is plotted for males and females.



(b)

—●— Male 20-29 Yrs Experience —◆— Female 20-29 Yrs Experience

Figure 21 Source: March CPS data for earnings years 1963-2008. See note to Fig. 19. Log college/high school weekly wage ratio for 0-9 and 20-29 years of potential experience is plotted for males and females.

An update of Card & Lemieux [01]

- Consider low (L) and high (H) skilled workers:

$$Y = [L^{\frac{\sigma_e-1}{\sigma_e}} + H^{\frac{\sigma_e-1}{\sigma_e}}]^{\frac{\sigma_e}{\sigma_e-1}}$$

with L and H aggregates across age groups j :

$$L \equiv [\sum_j L_j^{\frac{\sigma_a-1}{\sigma_a}}]^{\frac{\sigma_a}{\sigma_a-1}} \text{ and } H \equiv [\sum_j H_j^{\frac{\sigma_a-1}{\sigma_a}}]^{\frac{\sigma_a}{\sigma_a-1}}$$

- Using the first-order conditions gives:

$$\ln\left(\frac{w_{Hj}}{w_{Lj}}\right) = -\frac{1}{\sigma_a} \left[\ln\left(\frac{H_j}{L_j}\right) - \ln\left(\frac{H}{L}\right) \right] - \frac{1}{\sigma_e} \ln\left(\frac{H}{L}\right)$$

An update of Card & Lemieux [01]

- We had that (adding time subscripts t):

$$\ln\left(\frac{w_{Hjt}}{w_{Ljt}}\right) = -\frac{1}{\sigma_a} \left[\ln\left(\frac{H_{jt}}{L_{jt}}\right) - \ln\left(\frac{H_t}{L_t}\right) \right] - \frac{1}{\sigma_e} \ln\left(\frac{H_t}{L_t}\right)$$

- Adding a constant, a quadratic in time to capture SBTC, and a vector of age group main effects to capture persistent productivity differences between age groups:

$$\ln\left(\frac{w_{Hjt}}{w_{Ljt}}\right) = \beta_0 + \beta_1 \left[\ln\left(\frac{H_{jt}}{L_{jt}}\right) - \ln\left(\frac{H_t}{L_t}\right) \right] + \\ \beta_2 \ln\left(\frac{H_t}{L_t}\right) + \beta_3 t + \beta_4 t^2 + \delta_j + \eta_{jt}$$

Table 9 Regression models for the college/high school log wage gap by potential experience group, 1963-2008.

	Potential experience groups (years)				
	All	0-9	10-19	20-29	30-39
Own minus aggregate supply	-0.272 (0.025)	-0.441 (0.136)	-0.349 (0.095)	0.109 (0.079)	-0.085 (0.099)
Aggregate supply	-0.553 (0.082)	-0.668 (0.209)	-0.428 (0.142)	-0.343 (0.138)	-0.407 (0.141)
Time	0.027 (0.004)	0.035 (0.011)	0.016 (0.008)	0.015 (0.007)	0.020 (0.008)
Time ² /100	-0.010 (0.004)	-0.023 (0.011)	0.007 (0.008)	0.001 (0.007)	-0.008 (0.009)
Constant	-0.056 (0.085)	-0.118 (0.212)	0.120 (0.169)	0.138 (0.145)	0.018 (0.144)
Observations	184	46	46	46	46
R-squared	0.885	0.885	0.959	0.929	0.771

Source: March CPS data for earnings years 1963-2008. See notes to [Figs 2](#) and [19](#).

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Within-group wage inequality in the canonical model

- The canonical model can also generate within-group wage inequality if there is heterogeneity in efficiency units:

$$Y = \left[[A_L L]^{\frac{\sigma-1}{\sigma}} + [A_H H]^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} \quad (2)$$

with $L = \int_{i \in L} l_i di$ and $H = \int_{i \in H} h_i di$.

- Relative earnings of workers in the same group, say L, are:

$$\frac{W_i}{W_{i'}} = \frac{w_L l_i}{w_L l_{i'}} = \frac{l_i}{l_{i'}} \text{ for } i, i' \in L$$

- Canonical model can generate within-group wage inequality, but this inequality is independent of the skill premium.

Overall inequality in the canonical model

- Assume two observable skill groups of college (C) and non-college (N) workers.
- Assume that a fraction ϕ_c of college workers and a fraction ϕ_n of non-college workers are high-skilled (H) with the remaining fractions being low-skilled (L).
- Denote the true skill wage premium by:

$$\omega \equiv \frac{w_H}{w_L}$$

- Denote the observed college wage premium by:

$$\omega^c \equiv \frac{w_C}{w_N}$$

Overall inequality in the canonical model

- The observed college wage premium can be written as:

$$\omega^c = \frac{w_C}{w_N} = \frac{\phi_c w_H + [1 - \phi_c] w_L}{\phi_n w_H + [1 - \phi_n] w_L} = \frac{\phi_c \omega + [1 - \phi_c]}{\phi_n \omega + [1 - \phi_n]}$$

- If $\phi_n < \phi_c$, ω^c is increasing in ω .
- Denote the within-group skill premium by $\omega^{\text{within}} = \omega$.
- For constant ϕ_c and ϕ_n , ω^c , ω^{within} and ω will move together.
- The canonical model can predict changes in within-group and overall inequality (but not very intuitively).

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Directed technological change

- The canonical model assumes SBTC is a steady process that can be captured by a time trend.
- However, the speed of technological change can be endogenous and directed by the relative supply of high-skill workers.
- A more skilled workforce induces skill-biased technological change.
- Steady SBTC could result from a steady increase in the supply of skills (thus uniting the two parts of Tinbergen's race).

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Summary

- The canonical model provides a parsimonious framework for thinking about the changes skill premium and changes in overall wage inequality.
- The canonical model has been very successful empirically.
- It explains observed fluctuations in the skill premium by a secular increase in the relative demand for college workers due to SBTC combined with fluctuations in the growth of relative skill supplies.
- However, the canonical model falls short of explaining several recent trends in advanced labor markets.

Problem and puzzles for the canonical model

- It does not provide an explanation why wages of unskilled workers have fallen (because it assumes q-complementarity).
- It does not explain the phenomenon of wage and job polarization (why demand for unskilled relative to middle skilled workers has increased).
- It is at odds with the idea that technologies directly replace workers (automation instead of factor-augmenting technological change).
- It is silent about the recent fall in the labor share.
- It is not informative about the role of imperfections (because it is a simple supply-demand framework).